In problem session Friday:

checking existence of path of length k in poly time (1)

=>

~~find the longest path in poly time (2)~~

checking existence of size k independent set in poly time (3)

=> ~~finding largest IS in poly time (4)~~

checking solvability of CIRCUITSAT in poly time

=>

finding a satisfying assn for CIRCUITSAT in poly time

reduce problem A to problem B in polynomial time

if B solvable in poly time then A solvable in poly time

why this focus on poly time?

- poly time = constant doubling time
- historically/empirically

  big polynomial → small polynomial
- BIG MYSTERY! does polynomial verifiability imply polynomial solvability?

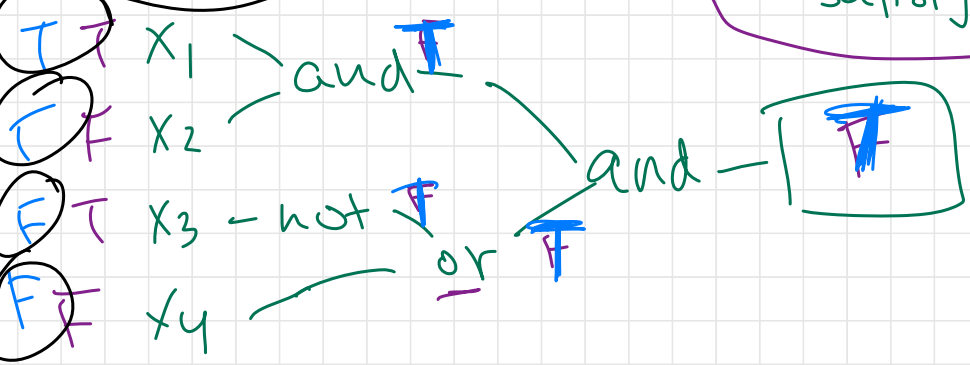
  P = NP?

NP = set of all decision problems whose "yes" answers are verifiable in poly time

Co-NP = set of all decision problems whose "no" answers are verifiable in poly time

Circuit SAT

not
and
or

$\bar{S}$

is there a satisfying assn?